Router, Permutation & MLA-RoPE Fusion

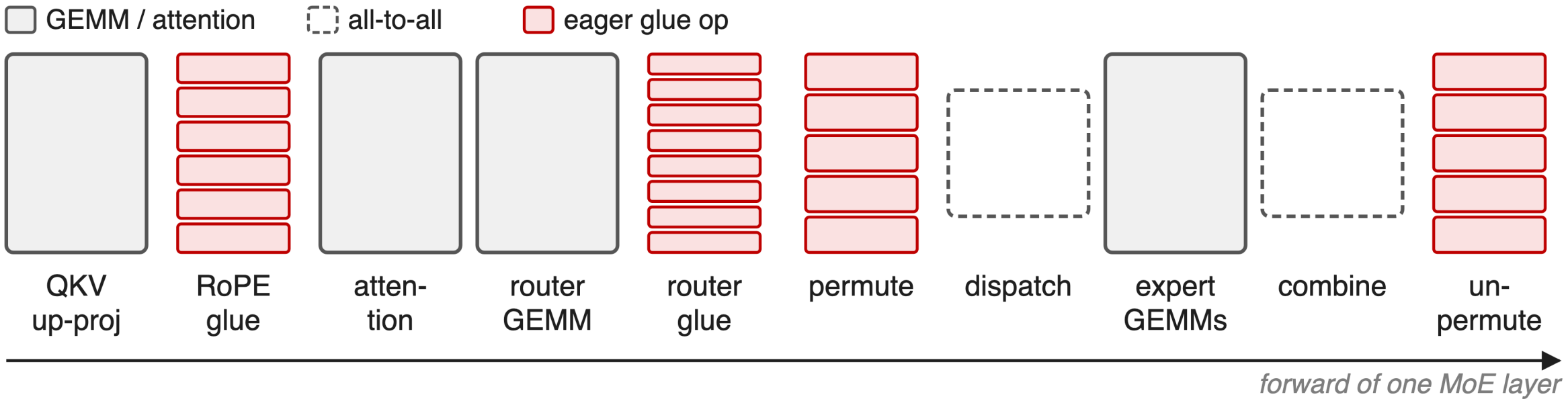
Collapse MoE and MLA glue ops into a few kernels

NVIDIA · Megatron-Core MoE · Transformer Engine

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Kernel Fusion & Runtime · arxiv.org/abs/2603.07685

Between the GEMMs, MoE layers run on glue



DeepSeek-V3: top-8 of 256 experts; RoPE on 64 of 192 dims per head

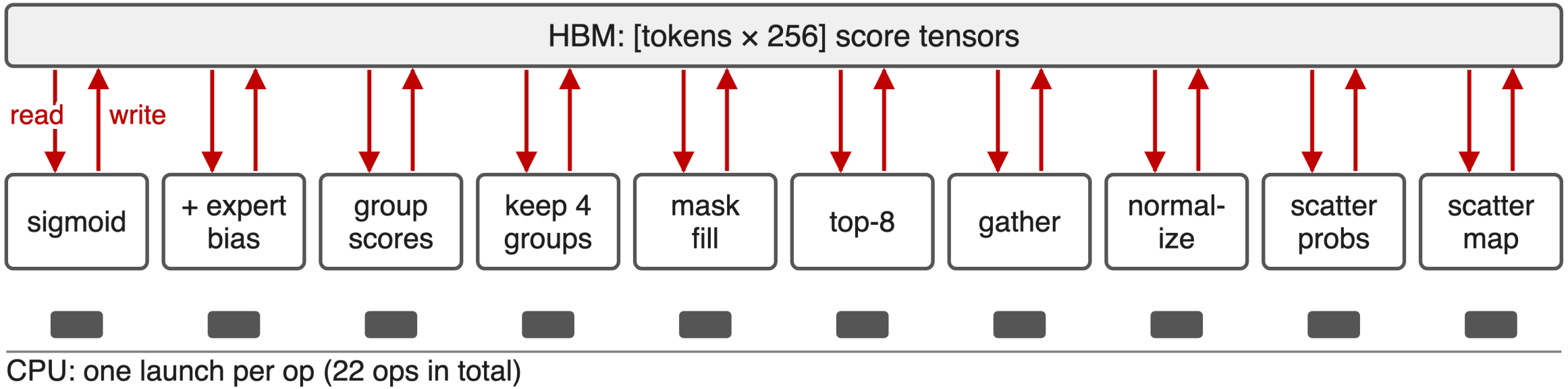
Routing one microbatch takes 22 eager ops

22

PyTorch ops in the unfused DeepSeek-V3-style router forward; × 58 MoE layers ≈ 1,276 launches per forward pass (derived)

Counted from Megatron-LM moe_utils.topk_routing_with_score_function, unfused path (derivation in notes/sources.md)

Every glue op re-streams the tensor through HBM



Each op is one launch plus a full read and write of a [tokens × 256] tensor

Graphs and per-op kernels only go halfway

- **CUDA graphs** (deck 38)
- One launch for many ops
- **Every HBM round trip remains**

- **Faster single ops**
- Each op still a full pass
- **Unpermute copies tokens to FP32**

Needed: run each per-token chain in one pass over the data

Insight: glue is per-token and fits on chip

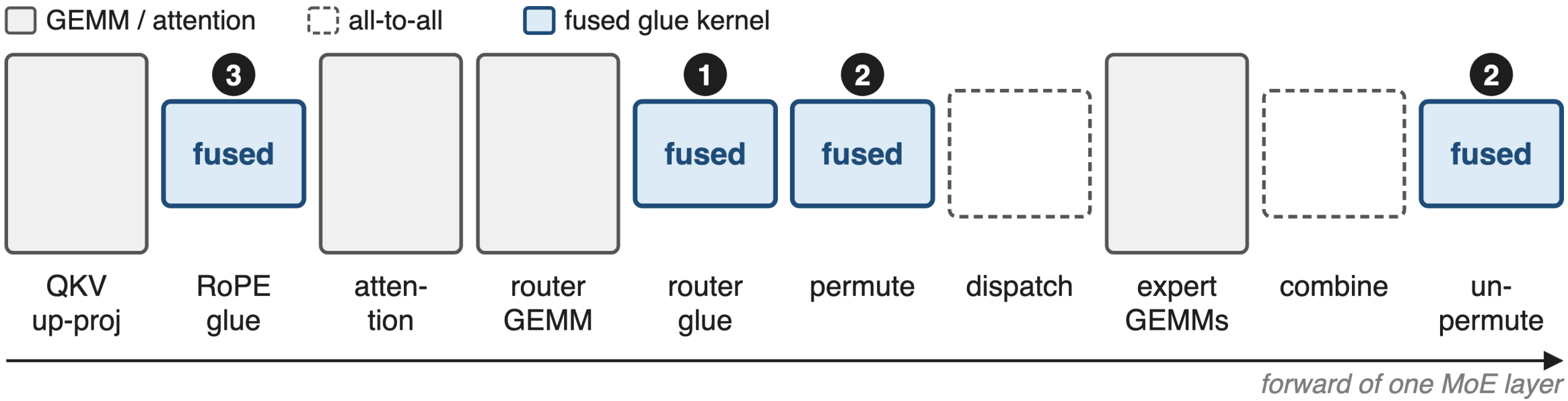
- Router: a token's 256 scores fit one warp's shared memory
- Permutation: one row-id map serves permute, unpermute, backward
- RoPE: only 64 of 192 dims per head change

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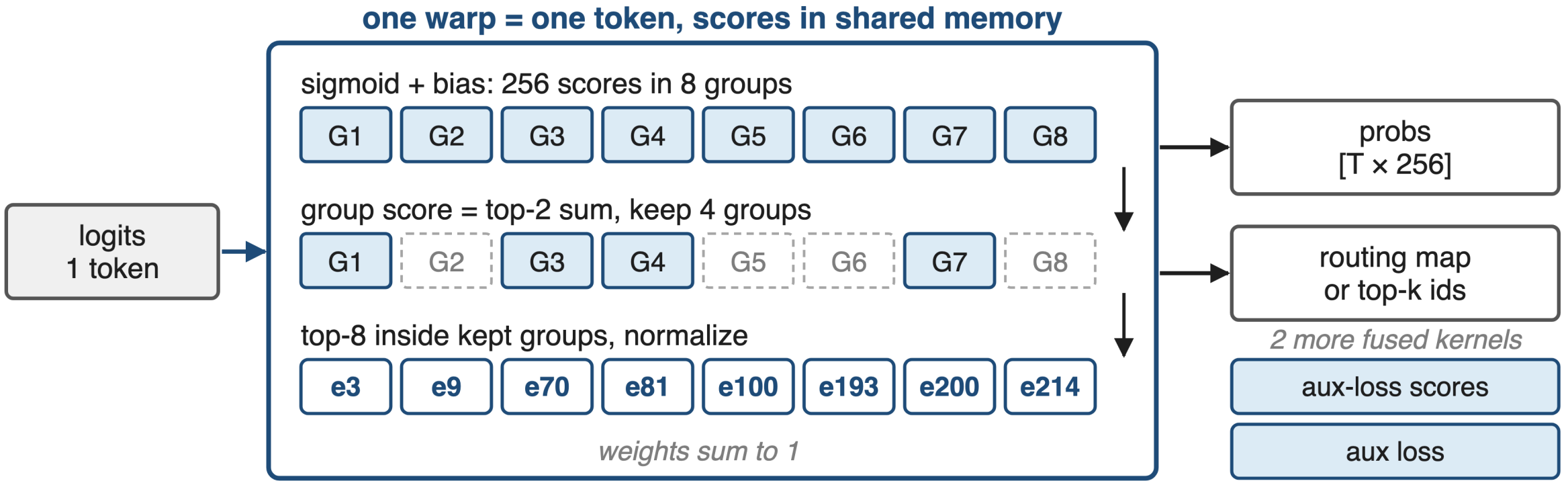
One kernel per chain: read once, compute on chip, write once

Three flags turn glue stacks into kernels



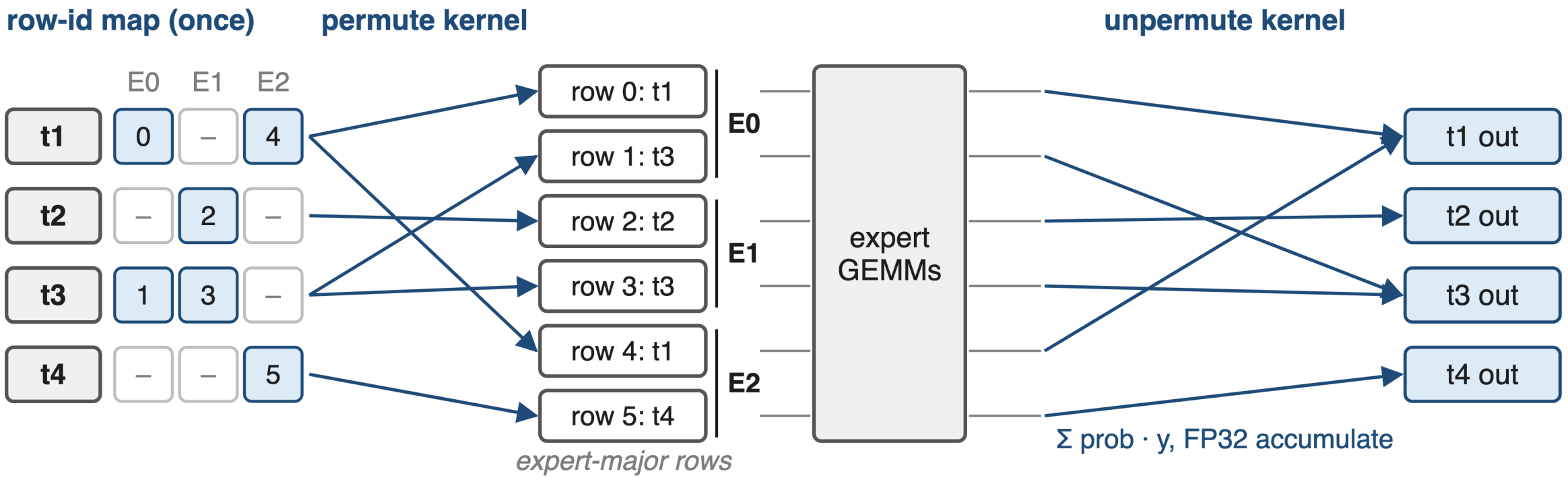
--moe-router-fusion (TE ≥ 2.7) · --moe-permute-fusion · --apply-rope-fusion (MLA with YaRN)

❶ One warp routes one token



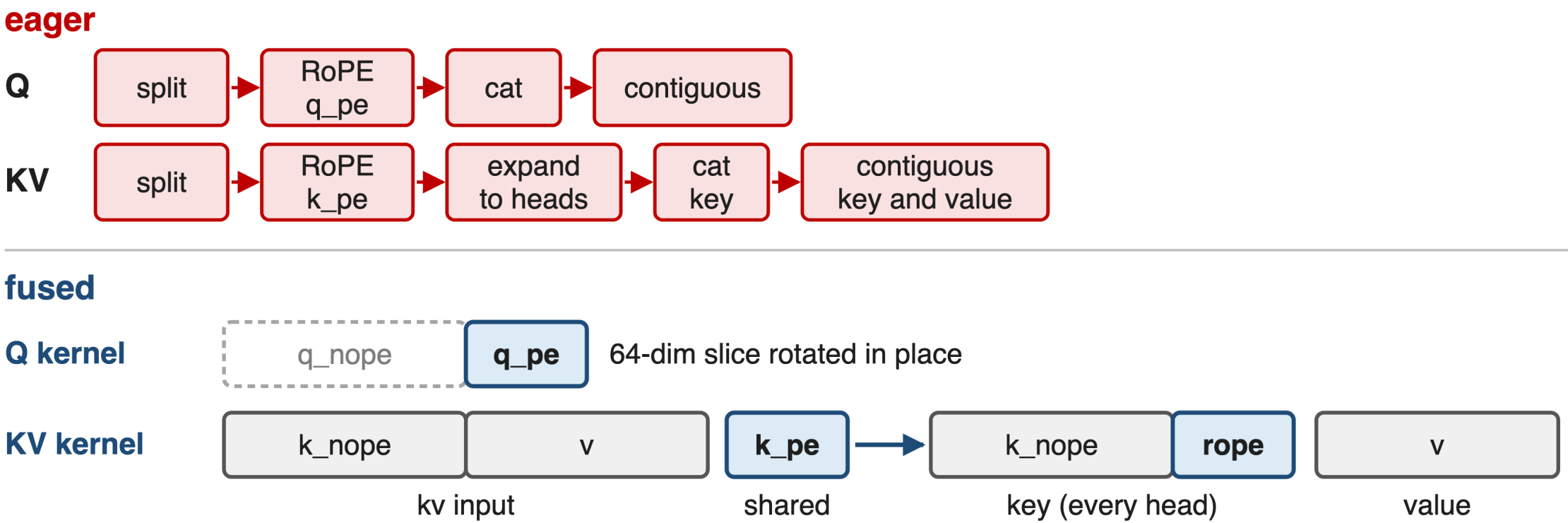
Backward fused too; the aux loss adds two fused kernels

❷ One row-id map drives permute and unpermute



Probs and FP8 scales move with the tokens (HybridEP fusion: deck 19)

❸ RoPE touches only the 64-dim slice

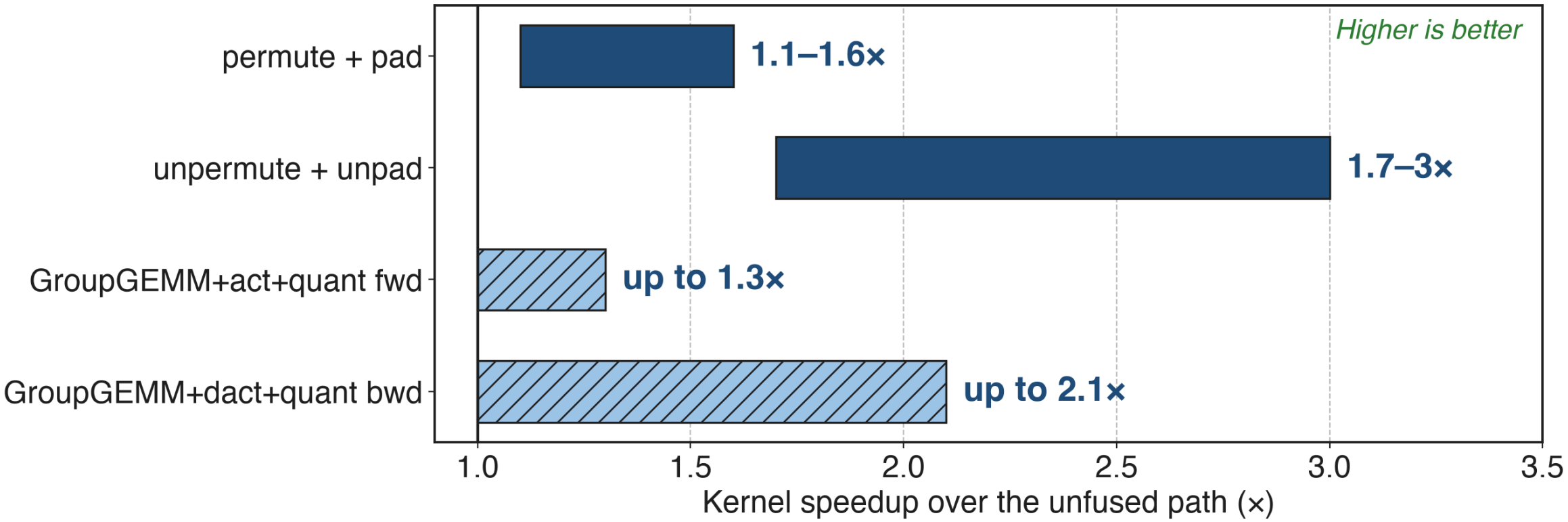


❹ Padding and quantization ride along

- Pad experts to 16/32/128 tokens inside the permute
- FP8 scaling factors permuted with their tokens
- Activation + quantize in the grouped-GEMM epilogue

Each fused step removes one more HBM pass, and stays CUDA-graph safe

Fused permutation kernels run 1.1–3× faster



Solid: measured ranges (TE PR #1921). Hatched: published maximum only (NVIDIA fusion-kernel blog)

Fusions add 0.4–8% on DeepSeek-class training

- Permute + pad: **+0.4%** MFU, ~**1 GB** less memory (~600B FP8 MoE)
- Fused expert kernels: up to **+8%** DeepSeek-V3, **+93%** GPT-OSS
- DeepSeek-V3 on GB200, all fusions + graphs: **1,048** TFLOPS/GPU

No isolated end-to-end number is published for router or RoPE fusion

Fusion trades determinism and flexibility

- **Fused aux loss not bitwise deterministic**
- Batch-invariant MoE needs unfused permute
- Quantile balancing: no router fusion
- MLA RoPE fusion: training only, YaRN only
- Needs TE ≥ 2.7 and Triton
- Each new router variant needs kernel work

Summary

- Eager router: **22** ops per layer (derived), each an HBM pass
- **Insight**: glue is per-token; one kernel per chain
- Router: one warp per token; permutation: one row-id map; RoPE in place
- **1.1–3×** fused permutation; **+8%** DeepSeek-V3 from fused experts

https://arxiv.org/abs/2603.07685